Primitive sets of permutations

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1 Introduction

A *primitive set* is a subset A of the positive integers where no element of A divides another (distinct) element. These are simply antichains of the poset $(\mathbb{N}, |)$, i.e. the positive integers with the divisibility partial order. Usually one imposes that $1 \notin A$, since if $1 \in A$ then necessarily $A = \{1\}$. Primitive sets have been studied since the 1930s, see the introduction to Lichtman [3]. Erdős, Sárközy and Szemerédi conjectured that [1],

$$\sum_{a \in A} \frac{1}{a \log a} \leq 1 + o(1)$$

holds uniformly over all the primitive sets $A \subseteq [x, \infty)$, as $x \to \infty$. This conjecture was recently settled by L. Price [4] via ChatGPT Pro. Motivated by the solution, and especially its distillation by T. Tao [5], we establish a permutation analogue. We first introduce a poset of permutations, whose definition may be of independent interest.

Definition 1. *Let $(S, \leq)$ be the following poset. The set S is the union $\cup_{n \geq 0} S_n$, where S_n is the symmetric group on $\{1, 2, \ldots, n\}$, and $S_0 = \{\varnothing\}$ is the singleton of the empty permutation. The relation $\pi \leq \sigma$ holds if $\pi \in S_m$ and $\sigma \in S_n$ for some $m \leq n$, and π is obtained from σ by deleting some cycles from the cycle decomposition of σ, whose total size is $n - m$, and then relabeling the remaining elements $i_1 < i_2 < \ldots < i_m$ as $1 < 2 < \ldots < m$.*

If $\pi \in S_n$ we write $|\pi|$ for n, i.e. the size of the domain of π. A *primitive set of permutations* is an antichain of $(S, \leq)$ that does not contain $\varnothing$.

Theorem 1. *Let A be a primitive set of permutations. Then*

$$\sum_{\pi \in A} \frac{1}{(|\pi| + 1)|\pi|!} \leq 1.$$

This is sharp, as demonstrated by taking $A = \cup_{n \geq 1} C(n)$ where $C(n)$ is the set of n-cycles in S_n. More generally, an equality is achieved for $A = \cup_{n \geq 1} C(n, k)$ where $C(n, k)$ is the set of permutations in S_n with k cycles. Indeed, since $\sum_{n \geq 1} |C(n, k)| x^n / n! = (-\log(1 - x))^k / k!$ for $|x| < 1$ [2, Eq. (7)], we find that

$$\sum_{\pi \in \cup_{n \geq 1} C(n,k)} \frac{1}{(|\pi| + 1)|\pi|!} = \sum_{n \geq 1} \frac{|C(n, k)|}{(n + 1)n!} = \int_0^1 \frac{(-\log(1 - x))^k}{k!} dx = \frac{1}{k!} \int_0^\infty e^{-t} t^k dt = 1.$$

In [2, Proposition 8], Gorodetsky, Lichtman and Wong observed that

$$\sum_{\pi \in \cup_{n \geq 1} C(n,k)} \frac{1}{|\pi||\pi|!} = \zeta(1 + k) > 1 + 2^{-k-1},$$

holds, but Theorem 1 suggests that $1/(|\pi||\pi|!)$ is not the right weight. With the weights of [2], we can obtain from Theorem 1 the following immediate corollary.

Corollary 1. *Let A be a primitive set of permutations contained in $\cup_{n\geq m} S_n$, $m \geq 1$. Then*

$$\sum_{\pi\in A} \frac{1}{|\pi||\pi|!} \leq 1 + \frac{1}{m}.$$

We briefly mention another famous question on primitive sets of integers. The Erdős primitive set conjecture, recently resolved by Lichtman [3], states that for primitive sets A, the series $\sum_{a\in A} 1/(a\log a)$ is maximized when A is the set of prime numbers. Since cycles are analogous to primes, one may view Theorem 1 as a permutation analogue of the Erdős primitive set conjecture, in addition to being analogous to the conjecture of Erdős, Sárközy and Szemerédi.

The proof of Theorem 1 and the particular poset were found by ChatGPT Plus. The proof relies, as did the distillation by Tao [5], on the Discrete divergence theorem which we now state.

Lemma 1 (Discrete divergence theorem [5])**.** *Let G be a directed weighted graph, with each edge e having a "flow" $w(e) \geq 0$, and only finitely many outgoing edges from any vertex. Define the "divergence" div_v at a vertex v to be the sum of the flows out of v minus the flows into v. Then, for any finite set Ω of vertices, the sum $\sum_{v\in\Omega} \mathrm{div}_v$ is equal to the outflow of Ω (the sum of all the flows from a vertex in Ω to a vertex outside of Ω) minus the inflow (the sum of all the flows from a vertex outside of Ω to one inside Ω).*

2 Proof of Theorem 1

Consider the directed graph whose vertices are elements of S, and where there is an edge $\pi \to \sigma$ whenever σ is obtained from π by deleting exactly one cycle. If $|\pi| = n \geq 1$ and the deleted cycle has length ℓ, assign that edge the flow

$$w(\pi \to \sigma) = \frac{\ell}{n(n+1)n!}.$$

We compute the inflow and outflow of nonempty permutations. Let $\pi \in S_n$, $n \geq 1$. The total outflow from π is

$$\mathrm{outflow}(\pi) = \sum_{C \text{ cycle of } \pi} \frac{|C|}{n(n+1)n!} = \frac{1}{(n+1)n!}.$$

We move on to the inflow. For each $\ell \geq 1$, we need to count the 'parents' of π in $S_{n+\ell}$, i.e. permutations $\sigma \in S_{n+\ell}$ obtained by adjoining an ℓ-cycle to π. To count them, we choose the ℓ new points added to π, and choose a cycle on them. This determines σ uniquely and gives

$$\binom{n+\ell}{\ell}(\ell-1)! = \frac{(n+\ell)!}{n!\,\ell}$$

parents. Each such parent contributes the flow

$$w(\sigma \to \pi) = \frac{\ell}{(n+\ell)(n+\ell+1)(n+\ell)!},$$

so the total inflow into π coming from parents of size $n+\ell$ is

$$\frac{(n+\ell)!}{n!\,\ell} \cdot \frac{\ell}{(n+\ell)(n+\ell+1)(n+\ell)!} = \frac{1}{n!(n+\ell)(n+\ell+1)}.$$

Summing over $\ell \geq 1$,

$$\mathrm{inflow}(\pi) = \frac{1}{n!}\sum_{\ell\geq 1} \frac{1}{(n+\ell)(n+\ell+1)} = \frac{1}{n!}\sum_{m\geq n+1}\left(\frac{1}{m} - \frac{1}{m+1}\right) = \frac{1}{(n+1)n!} = \mathrm{outflow}(\pi).$$

So for every $\pi \neq \varnothing$, the divergence at π is 0:

$$\mathrm{div}_\pi = 0.$$

Let A be an antichain of the poset S; we may assume A is finite. Let

$$\Omega := \{\varnothing \neq x \in S : \exists y \in A \text{ such that } x \leq y\}$$

be the set of all nonempty elements in S less than or equal to some element of A. If $\pi \in A$, then every parent of π lies outside Ω by the assumption that A is an antichain. Therefore the total inflow into Ω is

$$\text{inflow}(\Omega) \geq \sum_{\pi \in A} \frac{1}{(|\pi|+1)|\pi|!}.$$

The only way an edge can leave Ω is by landing at the empty permutation $\varnothing$. Thus the outflow of Ω is at most the total inflow into $\varnothing$. Only cycles contribute to $\varnothing$, so the total inflow into $\varnothing$ is

$$\text{inflow}(\varnothing) = \sum_{n \geq 1} (n-1)! \frac{1}{(n+1)n!} = 1.$$

Now we apply Lemma 1 to Ω. Since every $\pi \neq \varnothing$ has divergence 0,

$$1 \geq \text{outflow}(\Omega) = \text{inflow}(\Omega) \geq \sum_{\pi \in A} \frac{1}{(|\pi|+1)|\pi|!},$$

which concludes the proof.

References

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